

Thermal-neutron statistics for X17 production at n_TOF EAR2

What 10^9 fully simulated neutrons say about the sub-keV rate tables

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Internal note — informal. Drafted by Claude (Anthropic AI assistant) under D. Neff's direction; simulation, numbers and text reviewed by D. Neff.

Part I — Summary

The question

The X17 search at EAR2 produces its signal through radiative neutron capture: a slow neutron absorbed by ^3He forms $^4\text{He}^*$ at 20.6 MeV, and that de-excitation is the only doorway to X17 (or to the internal-pair background, IPC). The expected event rate per proton pulse therefore scales directly with the number of *radiative* captures per pulse. The collaboration rate table (`calculation_tables/results.3He`) quotes 1.2×10^{-2} IPC/pulse from neutrons below 1 keV — the number that anchored the $3.0\sigma/30$ -day sensitivity projection.

The problem

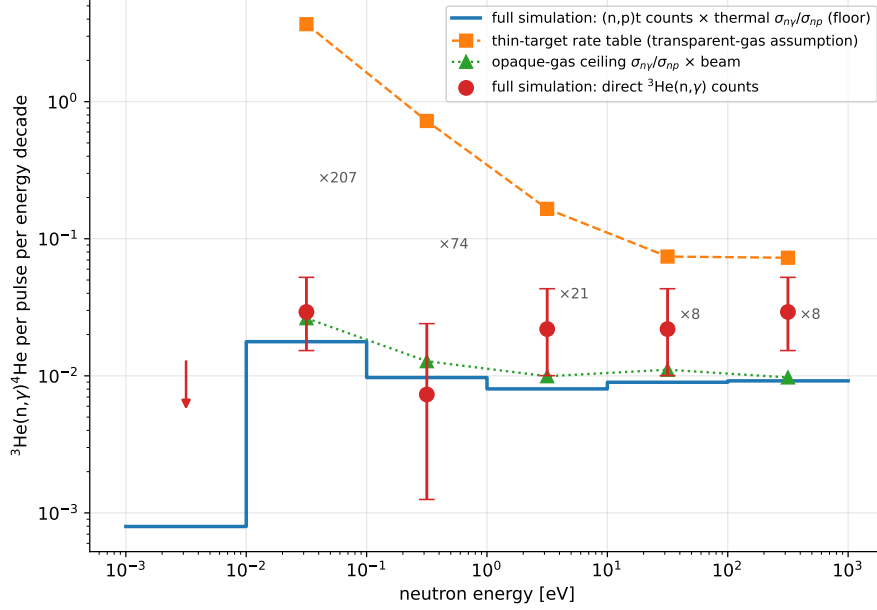
When a sub-keV neutron is absorbed by ^3He , two reactions compete: breakup, $^3\text{He}(n,p)t$ ($\sigma_{np} = 5333$ b at thermal), and radiative capture ($\sigma_{n\gamma} = 54 \mu\text{b}$). **Breakup wins about 10^8 times more often, and it is so strong that it consumes the entire beam** — a thermal neutron survives only ~ 0.15 mm into the 60 mm gas column. Once every neutron is absorbed, the number of radiative captures is set by the competition alone: per beam neutron it can never exceed $\sigma_{n\gamma}/\sigma_{np} \approx 1.0 \times 10^{-8}$, no matter how much gas sits behind the entrance face. The rate table instead multiplies $\sigma_{n\gamma}$ by the *full* gas column — counting radiative-capture opportunities in centimetres of gas that the (n,p) reaction never lets the neutrons reach. That overshoots the ceiling by a factor $\times 8$ to $\times 200$ depending on energy.

What we did

Run B of the Geant4 campaign fired 10^9 neutrons at the full STEP-derived capsule geometry, with energies drawn from the evaluated EAR2 flux and transverse positions from the energy-dependent beam profile, using high-precision neutron data (G4NDL). Self-shielding, scattering, beam halo and geometry are then automatic — no analytic assumption. Every neutron's terminal interaction (volume, process, position, energy) was recorded.

The answer

X17-production captures per pulse, sub-keV region
simulation total: $5.4\text{e-}02$ (floor) – $1.1\text{e-}01$ (direct) /pulse ($\rightarrow 1.1\text{e-}04$ – $2.3\text{e-}04$ IPC/pulse; table: 1.2×10^{-2})



The full simulation (blue curve, with the fifteen directly observed radiative captures as red points) lands on the opaque-gas ceiling (green) and confirms the rate table (orange) is too optimistic by $\times 8$ – 200 per decade. Folding in the beam spectrum and the n.TOF duty cycle (1.929×10^4 pulses/day at 7×10^{12} protons/pulse, from `results_3He`), all integrated over $E_n = 1\text{ meV} - 1\text{ keV}$:

$E_n = 1\text{ meV} - 1\text{ keV}$	per pulse		per day	
	rate table	simulation	rate table	simulation
radiative captures	5.8	$(0.5 - 1.1) \times 10^{-1}$	1.1×10^5	$(1.0 - 2.1) \times 10^3$
IPC pairs	1.21×10^{-2}	$(1.1 - 2.3) \times 10^{-4}$	233	2.2–4.4
X17 (2.5% of IPC)	—	$(2.9 - 5.7) \times 10^{-6}$	(5.8)	0.06–0.11

The bottom line is the lower-right cell: about one X17 produced every 9–18 days — before any detector acceptance, trigger or analysis efficiency. A 30-day sub-keV run would produce 2–3 X17 decays in total. The measurement is simply not feasible in this energy window, independent of any background consideration.

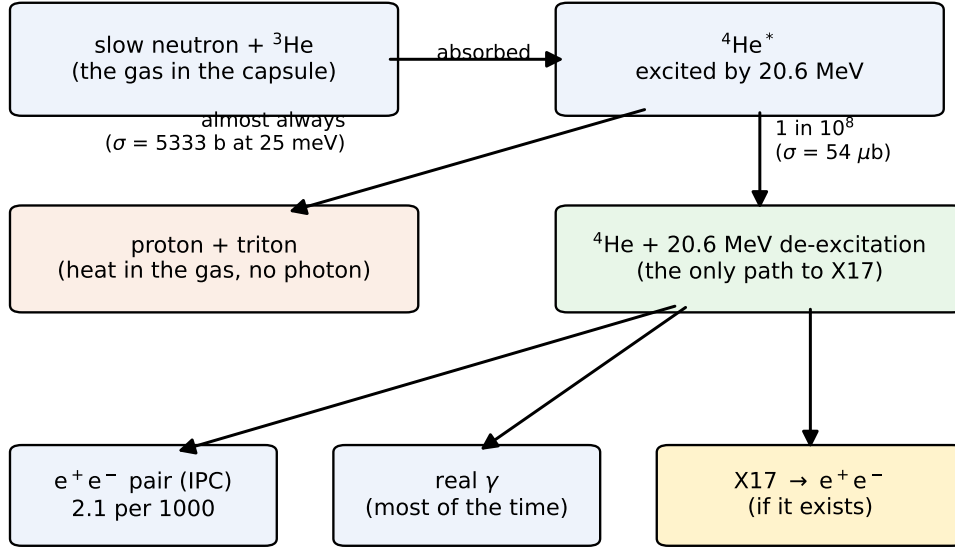
The sub-keV statistics are a factor 50–100 below the table. (The range spans two estimators: the thermal branching-ratio floor gives $\times 106$; the direct ${}^3\text{He}(n,\gamma)$ count — which includes the rise of $\sigma_{n\gamma}/\sigma_{np}$ toward 1 keV, Sec. 1 — gives $\times 53 \pm 14$.) If the sensitivity projection is anchored to the sub-keV rows, the $3.0\sigma/30\text{ d}$ becomes ~ 0.3 – $0.4\sigma/30\text{ d}$. The physical rate is not gone, however: above $\sim 100\text{ keV}$ the gas really is thin and the table is valid there; those rows carry $\sim 98\%$ of the total radiative captures. The likely conclusion is that **the analysis region of interest must move to the 0.1–10 MeV region**, which changes the generator kinematics and the gamma-flash/TOF situation — to be discussed with Alberto (run B-full, 1 meV–100 MeV, is already on disk for this).

Part II — The full story, from the beginning

1 What can happen when a slow neutron meets ^3He

When the neutron is absorbed, the compound nucleus $^4\text{He}^*$ holds 20.6 MeV of excitation energy. It almost always sheds it by breaking into a proton and a triton — a quiet outcome that deposits heat in the gas and produces no high-energy photon. About once in 10^8 absorptions it instead de-excites electromagnetically, and only this branch can produce a real 20.6 MeV photon, an internal e^+e^- pair, or — if it exists — the X17 boson. Counting X17 statistics is therefore exactly the business of counting these rare radiative captures.

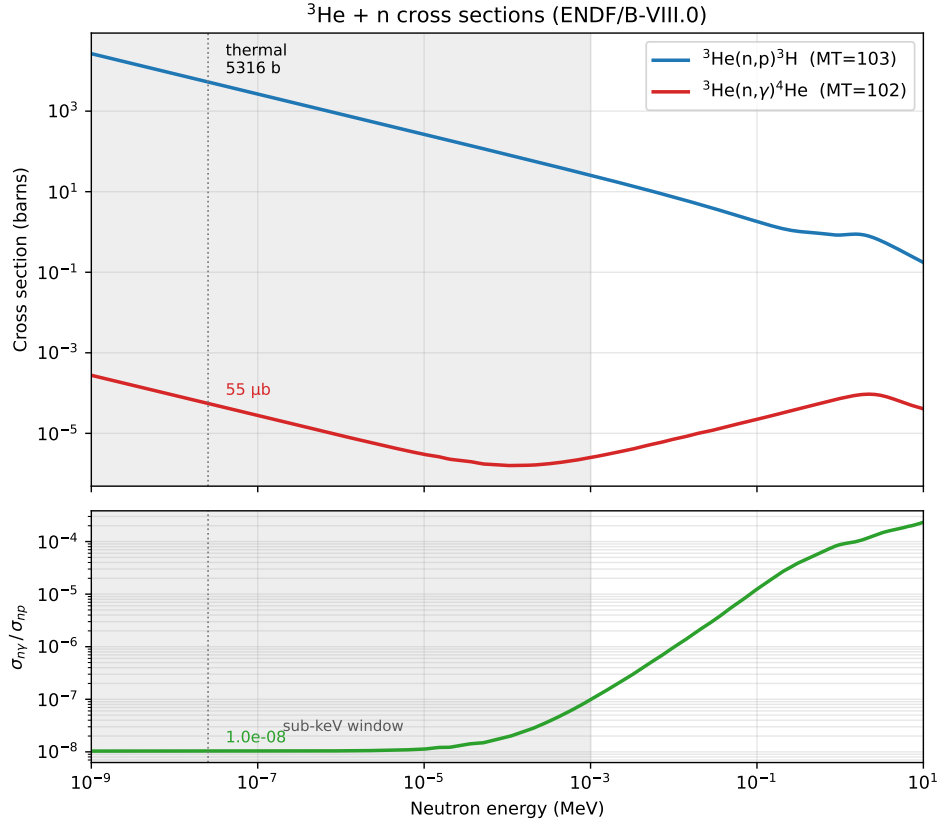
What can happen when a slow neutron meets ^3He



The two relevant cross-sections at thermal energy (25 meV): breakup $\sigma_{np} = 5333$ b, radiative $\sigma_{n\gamma} = 54 \mu\text{b}$ (Fig. below, ENDF/B-VIII.0). At thermal their ratio is

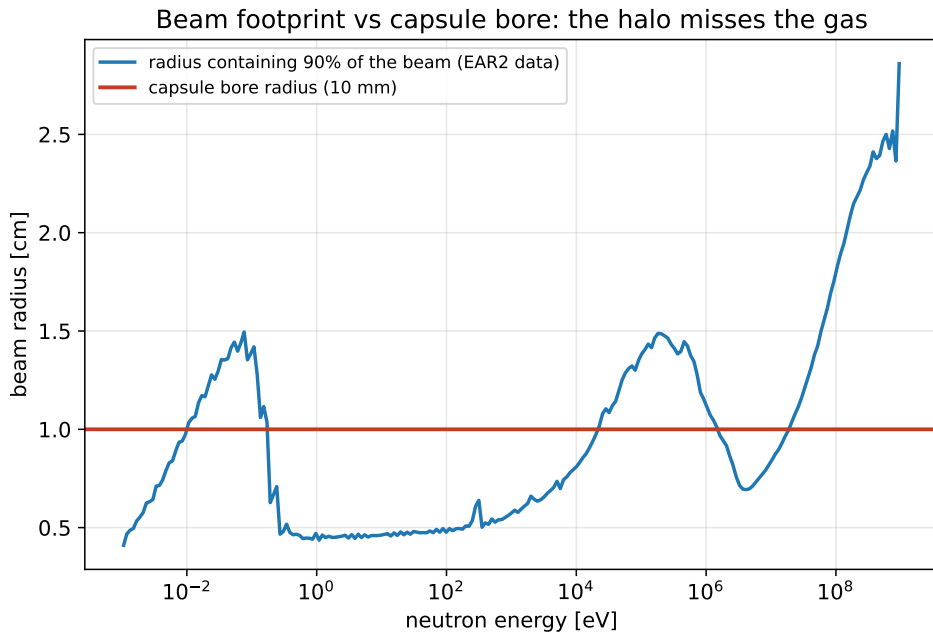
$$\frac{\sigma_{n\gamma}}{\sigma_{np}} = \frac{54 \times 10^{-6}}{5333} = 1.0 \times 10^{-8}. \quad (1)$$

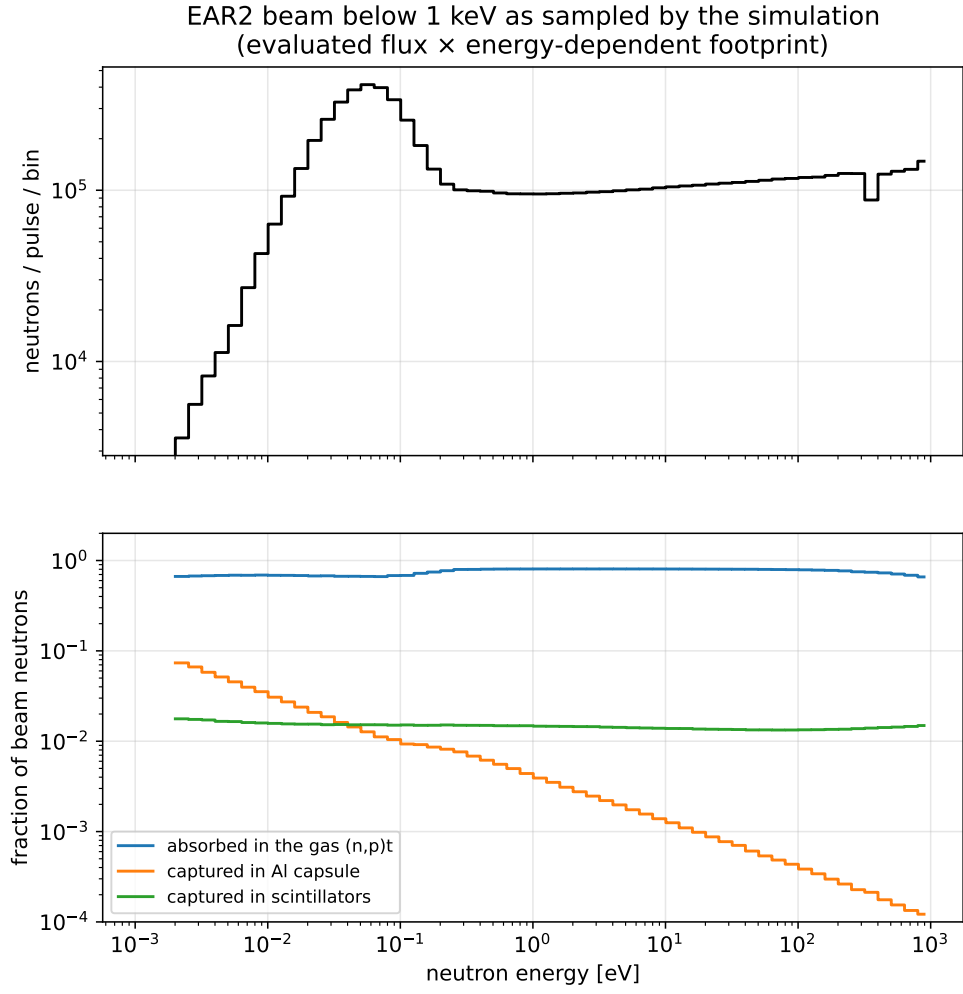
Both cross-sections fall as $1/v$ at low energy, so the ratio is constant below ~ 1 eV and rises only gently (to $\sim 2 \times 10^{-8}$) toward 1 keV, where $\sigma_{n\gamma}$ picks up a non- $1/v$ component — visible in the lower panel. Keep this number in mind: it is the punchline of the entire note.



2 What the beam delivers

EAR2 delivers 7.31×10^6 neutrons per proton pulse below 1 keV (evaluated Phase-3 flux). The beam is not a pencil: its transverse size grows toward low energies, while the capsule bore is only 20 mm across, so an energy-dependent fraction of the beam misses the gas entirely or hits the thick aluminium shoulder instead. The simulation samples both the energy spectrum and this energy-dependent footprint, so all acceptance effects are folded in from the start.



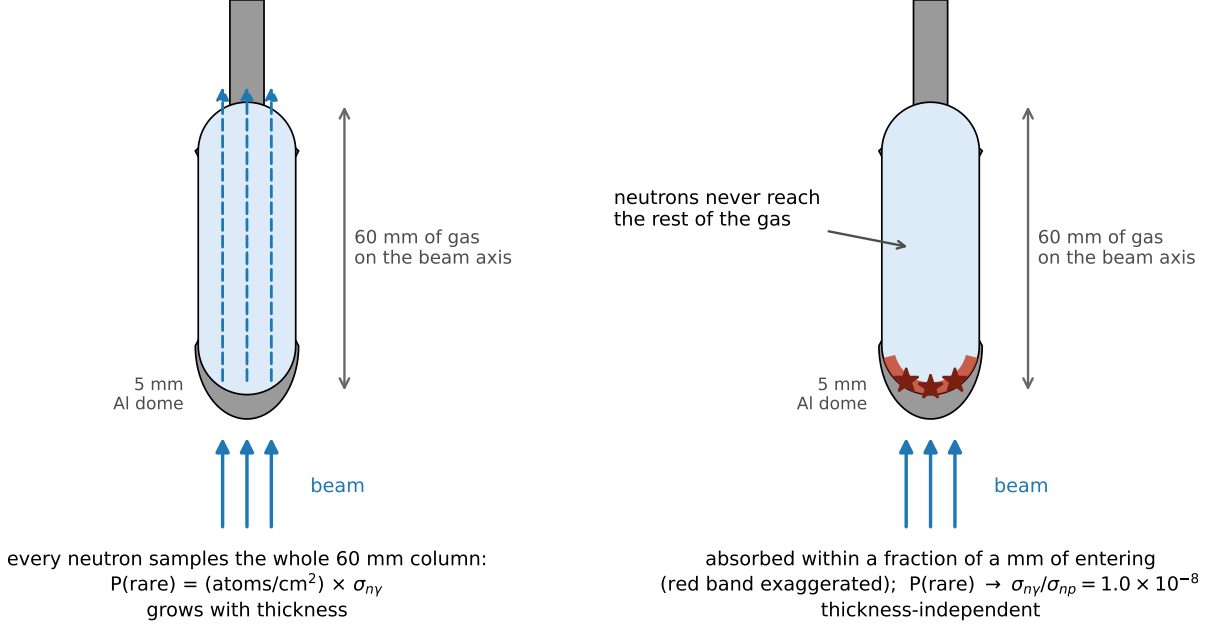


In numbers, for the sub-keV beam: 73.6% of all beam neutrons are absorbed in the gas, 23.9% leave the world without interacting (mostly footprint tail outside the bore), and the remainder is captured in surrounding material — of which more in Sec. 6.

3 The core issue: the gas is a black absorber, not a thin foil

The assumption: gas is transparent
(thin-target formula)

The reality below 1 keV: gas is opaque
(mean free path ≈ 0.15 mm at 25 meV)



The rate table computes, for each energy bin,

$$P_{\text{rad}} = nL \cdot \sigma_{n\gamma}(E), \quad (2)$$

with nL the gas column density. The table assumes $nL = 3.16 \times 10^{-2}$ atoms/b (its geometry was a 4 cm gas sphere); the real capsule presents an even thicker 60 mm on-axis column, $nL = 7.5 \times 10^{-2}$ atoms/b. This is the standard *thin-target* formula and it is correct *only* when the neutron actually traverses the full column, i.e. when $nL \sigma_{np} \ll 1$. At 25 meV, however, $nL \sigma_{np} \approx 150$ for the table's column (≈ 400 for the capsule on axis): a neutron penetrates a fraction of a millimetre before being absorbed. In that regime the probability that the absorption is radiative saturates at the branching ratio:

$$P_{\text{rad}} \longrightarrow \frac{\sigma_{n\gamma}}{\sigma_{np}} = 1.0 \times 10^{-8} \quad (\text{independent of } L). \quad (3)$$

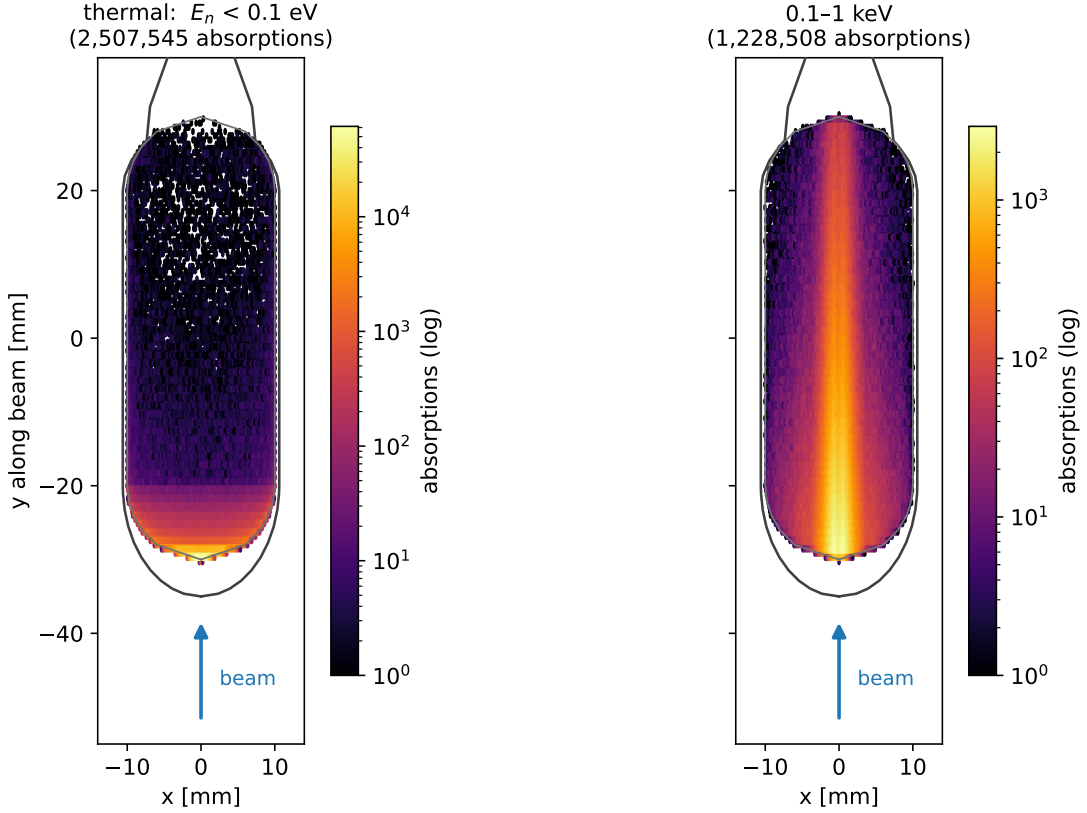
Dividing Eq. (2) by Eq. (3) shows the thin-target formula overshoots by exactly the opacity $nL \sigma_{np}$ of the column it assumed — for the table, a factor 150 at thermal, falling to ~ 7 at 1 keV. Adding more gas, or pressurising further, gains nothing below 1 keV: every neutron is already absorbed, and the radiative branch can only ever win its fixed share.

To be clear about what matters here: the point is *not* where in the gas the absorptions happen — it is that the (n,p) reaction destroys the beam before the rest of the gas can contribute. The absorption positions are simply the visible proof of that destruction, which is why we show them next.

The cartoon is not an exaggeration. Below is the same picture taken from the simulation itself: the measured absorption positions of beam neutrons in the gas, drawn inside the true capsule profile. At thermal energies the absorptions pile up in a shell hugging the inside of the entrance hemisphere; the interior sees 10^3 – 10^4 times fewer (the faint fill, and the bright rim along the lower side walls, are neutrons that entered through the thin 0.6 mm barrel — the thermal

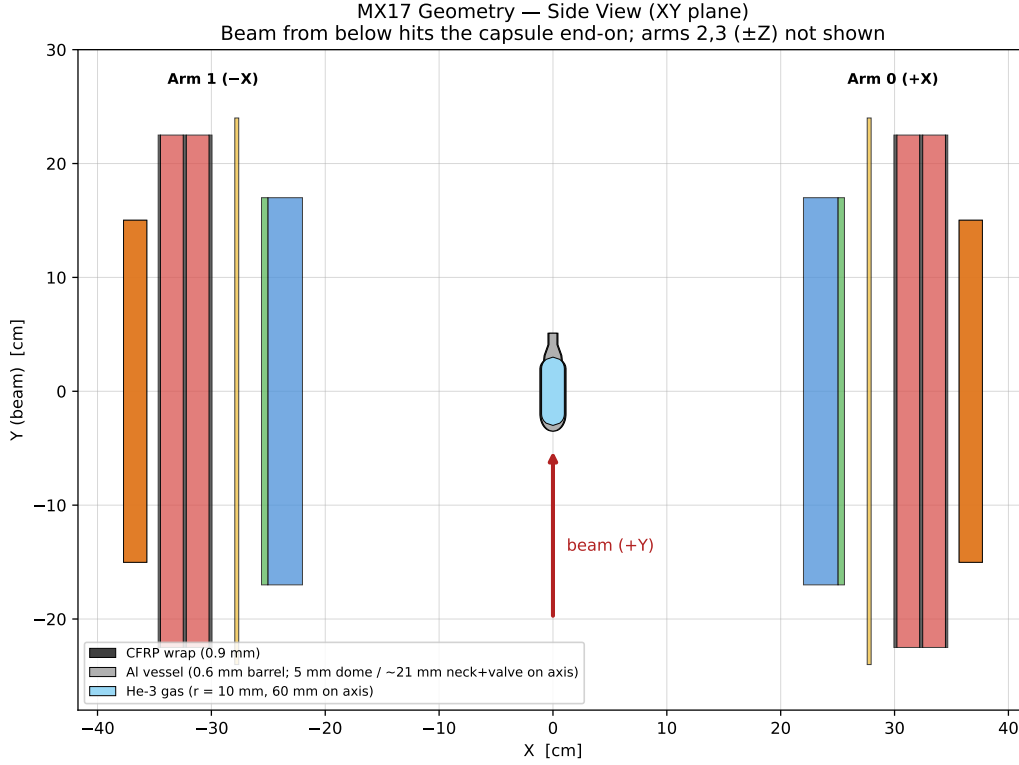
beam is wider than the bore — or scattered before being absorbed). By 0.1–1 keV the mean free path has grown to centimetres and the absorptions form a column through the whole volume, still thinning toward the top as the column self-attenuates.

Where beam neutrons are actually absorbed in the gas (simulation, one job file)



4 What the full simulation adds

The cartoon argument above is one-dimensional. The real situation (Fig. below) has the beam arriving from below along the capsule axis, entering through a 5 mm-thick domed aluminium tip; the gas column length depends on where the neutron enters; the beam is wider than the bore; elastic scattering can bounce neutrons around; and the four detector arms surround the capsule transversely at 22 cm. Geant4 with high-precision neutron data (G4NDL) transports every neutron through the actual STEP-derived geometry and lets nature take its course. The simulation records, for each of the 10^9 neutrons of run B, where its track ended, by which process, and at what energy — so the capture accounting is exact, including all geometry and transport effects.

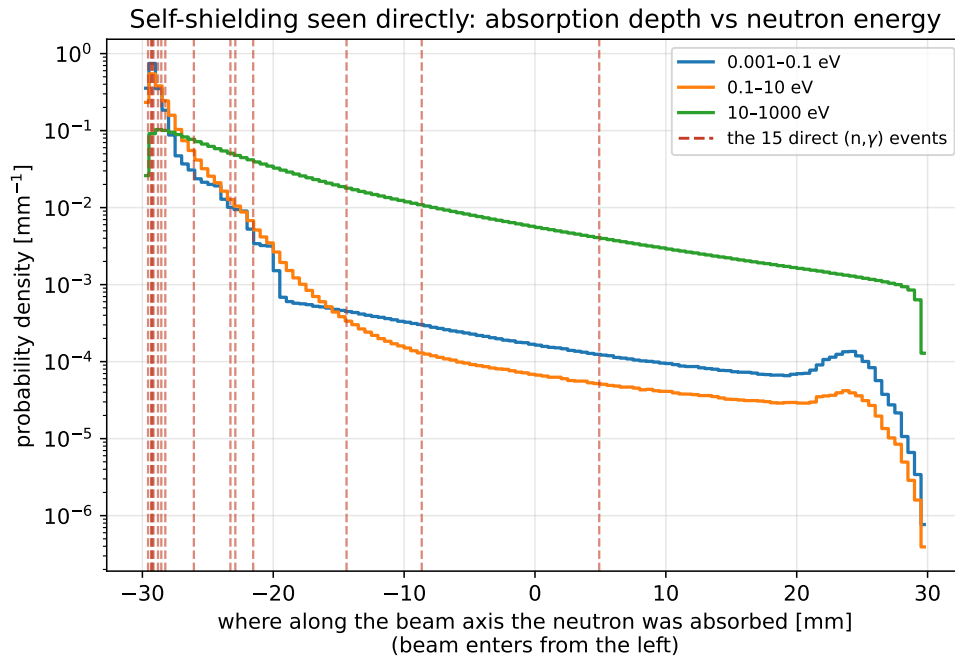


Capsule dimensions (STEP-derived): gas bore $r = 10$ mm, 60 mm long on axis (40 mm cylinder + hemispherical ends); Al wall 0.6 mm on the barrel but **5 mm on axis upstream** (dome) and **~21 mm on axis downstream** (neck + valve), plus a 0.9 mm CFRP overwrap.

5 The evidence

5.1 Self-shielding, observed directly

If the opaque-gas picture is right, thermal neutrons must pile up at the entrance face of the gas while keV neutrons penetrate deeper. That is exactly what the absorption-depth distribution shows:



Below 0.1 eV the absorption depth is a sliver at the entrance; by 0.1–1 keV it spreads through the column. The fifteen directly observed radiative captures follow the same pattern: the eleven below 100 eV all sit within a few millimetres of the entrance window, while the four in the 0.1–1 keV decade are distributed through the gas (full list in Appendix B).

5.2 The rate, measured three ways

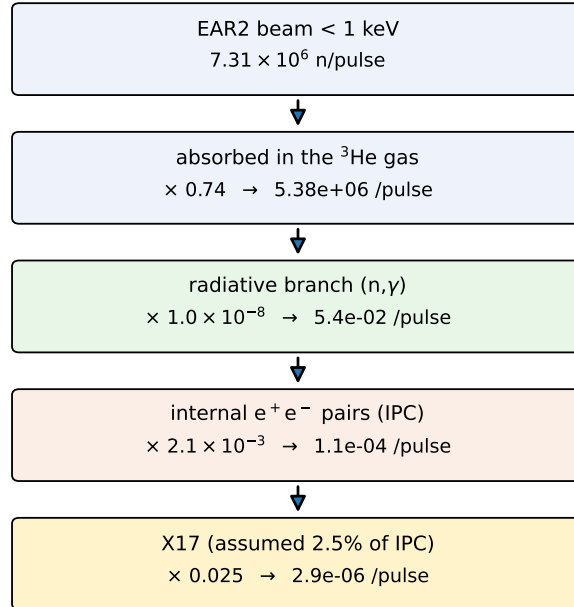
The headline figure (Part I) compares, per energy decade:

1. **Direct counts:** fifteen ${}^3\text{He}(n,\gamma)$ events in 10^9 neutrons $\Rightarrow (1.5 \pm 0.4) \times 10^{-8}$ per beam neutron.
2. **Branching-ratio floor:** 7.36×10^8 observed (n,p)t absorptions multiplied by the *thermal* branching ratio of Eq. (1) — exact below ~ 1 eV where the ratio is flat, a mild underestimate above (the ratio roughly doubles by 1 keV; Sec. 1).
3. **The analytic ceiling:** Eq. (3) times the beam.

All three agree within the expected energy dependence; the rate table exceeds them by $\times 200$ (thermal decade) down to $\times 2$ –8 (100 eV–1 keV); the row-by-row comparison is in Appendix A. As a by-product, the direct counts test the G4NDL nuclear data: the overall ratio $N_{n\gamma}/N_{np} = (2.0 \pm 0.5) \times 10^{-8}$, with the thermal events alone consistent with 1.0×10^{-8} and the eV–keV events sitting at the elevated ENDF ratio, exactly as the cross-section figure predicts.

5.3 From captures to X17 per pulse

Per-pulse normalisation ladder (sub-keV, full simulation)



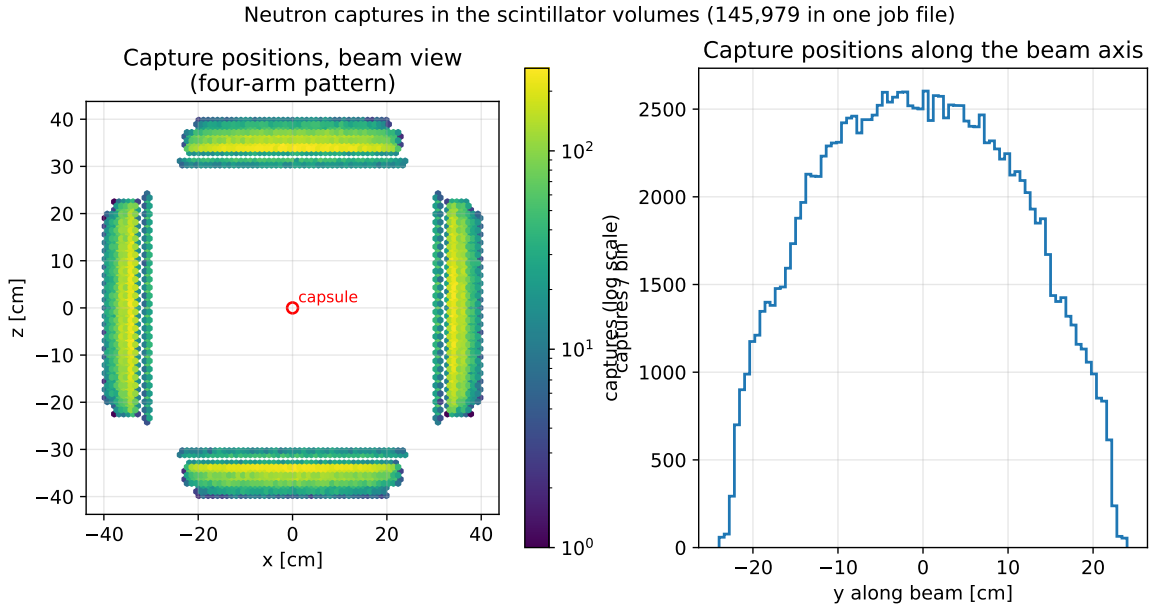
6 Where the neutrons actually end up

A secondary deliverable of run B, relevant for backgrounds rather than signal statistics. Per beam neutron below 1 keV:

fate	per beam neutron	comment
$^3\text{He}(n,p)t$ in gas	7.4×10^{-1}	signal-region heat, no photons
escapes world	2.4×10^{-1}	footprint tail
capture in Al capsule	8.0×10^{-3}	7.7 MeV γ cascades near target
capture in liquid scint.	1.2×10^{-2}	2.2 MeV γ born <i>inside</i> calorimeter
capture in plastic scint.	2.2×10^{-3}	trigger + back layers
capture in CFRP/PCB/mesh	$\sim 1.5 \times 10^{-3}$	distributed
$^3\text{He}(n,\gamma)$	1.0×10^{-8}	the X17 channel

Two observations. First, the aluminium capture rate is ~ 60 times the old flat-cap toy estimate — the STEP capsule’s domed tip and thick shoulder put far more Al in the beam. Second, and not in any toy so far: **hydrogen capture in the liquid scintillator is the largest single capture channel outside the gas**, and its 2.22 MeV photon is born inside the calorimeter itself. Both feed the wall-background simulation (run C) via the capture-vertex library.

How do beam neutrons end up in detectors 25 cm off to the side? They do not fly there directly — the beam points up, the arms sit sideways. But a neutron is not absorbed on its first encounter with aluminium: it usually just ricochets. Beam-halo neutrons that miss the 20 mm bore strike the thick Al dome, shoulder or valve, scatter to large angles, and then wander through the air. The four arms (each $\sim 40 \times 40$ cm at 22–31 cm) cover a large fraction of the sky as seen from the capsule, and the LAB liquid scintillator is hydrogen-rich — for a slow neutron, hydrogen is both an efficient brake and an efficient absorber. The capture positions confirm this picture: they reproduce the four-arm cross pattern around the capsule.



7 What this means for the experiment

- **Sub-keV ROI:** $(1.1\text{--}2.3) \times 10^{-4}$ IPC/pulse and $(2.9\text{--}5.7) \times 10^{-6}$ X17/pulse (at the assumed 2.5% of IPC pairs) — a factor 50–100 below the table. A thermal-only analysis cannot reach the projected significance; scaled naively, $3.0\sigma \rightarrow 0.3\text{--}0.4\sigma$ over 30 days.
- **The MeV region:** above ~ 100 keV the gas opacity drops below 5% and the thin-target rows of the table become valid; they carry $\sim 98\%$ of all radiative captures. The commented-out 0.2–2 MeV row (5.2×10^{-2} IPC/pulse) looks like the right anchor. Moving there requires

(a) energy-dependent pair kinematics in the generator ($E^* \approx 20.58 \text{ MeV} + \frac{3}{4}E_n$ plus a centre-of-mass boost), and (b) an assessment of the gamma flash and TOF window at μs timescales. Run B-full (1 meV–100 MeV) is on disk to quantify the capture budget there.

- **For discussion with Alberto:** which rows anchored the sensitivity; the provenance of the IPC coefficient (2.1×10^{-3} — multipolarity, E0 contribution); and the capsule-geometry update (the table assumed a 4 cm sphere).

Caveats

- The G4NDL ${}^3\text{He}(n,\gamma)$ strength is validated here at the $\sim 25\%$ level (15 events); the branching-ratio argument does not depend on it.
- Room-return neutrons and upstream beamline structures are not in the simulation; they affect backgrounds, not the signal-rate conclusion.
- The simulated radial profile and flux are the evaluated Phase-3 EAR2 ones; if the beam group provides updated halo measurements, the footprint losses (24% escape) could shift.

A Row-by-row comparison with the rate table

Radiative captures per *beam* neutron, per energy decade. “Table” reproduces `results_3He` (thin-target); “ceiling” is Eq. (3) corrected for the (n,p) opacity within the decade; “floor” is the (n,p)×thermal-BR measurement from 10^9 neutrons (statistical errors negligible, but an underestimate above ~ 1 eV where the ENDF ratio rises); “direct” are the raw $^3\text{He}(n,\gamma)$ measurements (Poisson errors).

E_n range	table	floor	direct	table/floor	table/direct	ceiling
1–10 meV	—	6.9×10^{-9}	0 obs.	—	—	—
10–100 meV	1.41×10^{-6}	6.8×10^{-9}	1.1×10^{-8} (4)	207	126	1.01×10^{-8}
0.1–1 eV	5.71×10^{-7}	7.7×10^{-9}	5.8×10^{-9} (1)	74	99	1.01×10^{-8}
1–10 eV	1.68×10^{-7}	8.2×10^{-9}	2.2×10^{-8} (3)	21	7.5	1.01×10^{-8}
10–100 eV	6.69×10^{-8}	8.1×10^{-9}	2.0×10^{-8} (3)	8.3	3.4	1.00×10^{-8}
0.1–1 keV	5.90×10^{-8}	7.5×10^{-9}	2.4×10^{-8} (4)	7.9	2.5	7.9×10^{-9}

The floor sits 20–30% *below* the ceiling because the ceiling assumes every beam neutron is absorbed in the gas, while in reality 67–81% are (footprint tail and window transmission). The direct measurements exceed the floor above 1 eV by the rising ENDF $\sigma_{n\gamma}/\sigma_{np}$ ratio (cross-section figure, Sec. 1) — the table/direct factors at the top of the window are correspondingly milder. Flux-weighted over the full sub-keV window the table gives 1.21×10^{-2} IPC/pulse against the simulation’s 1.1×10^{-4} (floor) to 2.3×10^{-4} (direct): **factor 50–100**.

B The fifteen direct events, and provenance

E_n	depth along beam	E_n	depth along beam
12.4 meV	−29.2 mm	8.11 eV	−29.6 mm
13.5 meV	−29.3 mm	15.7 eV	−28.8 mm
41.9 meV	−29.3 mm	17.3 eV	−21.5 mm
64.4 meV	−29.2 mm	80.9 eV	−26.1 mm
0.735 eV	−28.3 mm	479 eV	+4.9 mm
3.27 eV	−28.6 mm	498 eV	−14.4 mm
4.14 eV	−23.3 mm	519 eV	−22.9 mm
		897 eV	−8.7 mm

The gas entrance face is at −30 mm: every event below 100 eV is absorbed within the first few millimetres; only the 0.1–1 keV events reach the interior.

Data: run B sub-keV, 100 validated jobs (10^9 neutrons),

`/eos/experiment/ntof/data/x17/full_sim/neutrons_subkev/`.

Scan: `scripts/analyze_thermal_captures.py` → `thermal_captures_subkev_full.npz/json`.

Figures: `scripts/make_thermal_report_figures.py`, `scripts/plot_he3_xs.py` (ENDF/B-VIII.0 via `data/He3.h5`). Repository: `MX17_Full_Geant`, branch `main`.